Chapter 2

Random Variables: General Theory

In this chapter we introduce **random variables**. In most situations and applications you will be working with random variables rather than with the underlying probability model for a random experiment. It is a peculiarity of the terminology used in probability theory that a random variable is actually a function! Here is the formal definition of a random variable:

Definition 2.1 (Random variable)**.** Suppose $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability model for a random experiment, and suppose that $X : \Omega \to \mathbb{R}$ is a *function* which assigns a numerical value $X(\omega)$ to every outcome $\omega \in \Omega$ such that

$$\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}, \quad \text{for any } x \in \mathbb{R}, \tag{2.1}$$

then we say that X is a random variable.

Remark 2.1.
- When working with random variables we often use the shorthand for any $A \subseteq \mathbb{R}$ to write
  $\{X \in A\} = \{\omega \in \Omega : X(\omega) \in A\}$.
- We will often consider sets of the form $\{a \leq X \leq b\}$ and $\{X \leq x\}$. Note that these are subsets of Ω and we wish to assign probabilities to these sets, the condition (2.1) states that $\{X \leq x\}$ is an event and hence we can assign a probability to this set.

Example 2.1. Let $\Omega = \{\omega_0, \omega_1\}$ with $\mathcal{F}$ the set of all subsets of Ω, i.e. $\mathcal{F} = \{\emptyset, \{\omega_0\}, \{\omega_1\}, \Omega\}$. Define $X : \Omega \to \mathbb{R}$ by $X(\omega_0) = 0$ and $X(\omega_1) = 1$. This is an example of the simplest non-trivial random variable, it takes the value 1 with probability $p = \mathbb{P}(\omega_1)$ and takes the value 0 with probability $1 - p$. This is known as a Bernoulli Random variable.

This can be used to model a coin flip, let $\omega_0 = T$ be the outcome of seeing "tails" and $\omega_1 = H$ be the outcome of seeing "heads". Then X counts how many heads are seen in one flip of the coin.

Definition 2.2 (Image of a random variable)**.** Suppose $X : \Omega \to \mathbb{R}$ is a random variable. Then the image of X (sometimes called the support of X), denoted by $\mathrm{Im}X$, is defined as

$$\mathrm{Im}(X) = \{x \in \mathbb{R} : X(\omega) = x \text{ for some } w \in \Omega\} = \text{set of all possible values taken by } X$$

Definition 2.3 (Discrete random variables)**.** Suppose that $X : \Omega \to \mathbb{R}$ is a random variable. We say that X is a discrete random variable if and only if $\mathrm{Im}X$ is a countable set.

Example 2.2 (Example 2.1 continued)**.** The image of X is $\mathrm{Im}(X) = \{0, 1\}$. This set is finite and hence countable, therefore X is a discrete random variable.

Example 2.3. Consider repeatedly flipping a coin until it shows "Heads". Let $X : \Omega \to \mathbb{R}$ denote the number of flips until "Heads" appears for the first time.

If we want to construct the probability space we could set

$$\Omega = \{\omega = (\omega_i)_{i=1}^n : n \in \mathbb{N}, \omega_i = T \text{ for } i < n, \omega_n = H\}.$$

that is, the set of all finite sequences where all but the last entry is T and the last entry is H. Then we can define $X(\omega)$ to be the length of the sequence ω, representing the number of flips until the first "Heads". Alternatively, we could set $\Omega = \mathbb{N}$ where $\omega \in \Omega$ represents the number of flips until the first "Heads" appears, in which case $X(\omega) = \omega$. In both cases, X is a discrete random variable since the image of X is $\mathrm{Im}(X) = \mathbb{N}$ which is countable.

Example 2.4. Let X denote the amount of time waiting at a bus stop until the next bus appears. Then the image of X is $[0, \infty)$ which is uncountable, hence X is not a discrete random variable.

2.1 Probability mass function (pmf)

Suppose $X : \Omega \to \mathbb{R}$ is a discrete random variable (i.e. $\mathrm{Im}(X)$ is countable). We define the probability mass function $p_X : \mathrm{Im}X \to [0, 1]$ by

$$\boxed{p_X(x) = \mathbb{P}(X = x) = \mathbb{P}\left(\underbrace{\{\omega : X(\omega) = x\}}_{\text{event in } \Omega}\right)}$$

Example 2.5 (Example 1.7 continued.)**.** $X = \#$heads in 3 flips of a fair coin. $\mathrm{Im}(X) = \{0, 1, 2, 3\}$. Probability mass function is

$p_X(0) = \mathbb{P}(X = 0) = \mathbb{P}\big(\{\mathrm{TTT}\}\big) = 1/8$
$p_X(1) = \mathbb{P}(X = 1) = \mathbb{P}\big(\{\mathrm{HTT}, \mathrm{THT}, \mathrm{TTH}\}\big) = 3/8$
$p_X(2) = \mathbb{P}(X = 2) = \mathbb{P}\big(\{\mathrm{THH}, \mathrm{HTH}, \mathrm{HHT}\}\big) = 3/8$
$p_X(3) = \mathbb{P}(X = 3) = \mathbb{P}\big(\{\mathrm{HHH}\}\big) = 1/8$

Note: In the notation $\mathbb{P}(X = x)$, notice that X and x are very different things.
X is the name of the random variable, and doesn't change so long as we're considering the same random variable.
Whereas x is a particular value that the random variable can take, and varies through all values in $\mathrm{Im}(X)$, so $x = 0, 1, 2, 3$ in Example 2.5 above.

Properties of probability mass functions

Suppose that p_X is the probability mass function of a (discrete) random variable X. Then:

1. $\forall x$, $p_X(x) \geq 0$
2. $\sum_x p_X(x) = 1$

Conversely, any function satisfying the above properties is a valid probability mass function for some discrete random variable.

Definition 2.4 (Cumulative distribution function)**.** Suppose $X : \Omega \to \mathbb{R}$ is a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$.
Then the cumulative distribution function (cdf) $F_X : \mathbb{R} \to [0, 1]$ is defined by

$$F_X(x) = \mathbb{P}(X \leqslant x) = \mathbb{P}\big(\{\omega : X(\omega) \leqslant x\}\big) \quad \forall x \in \mathbb{R}$$

Remark: *Every* random variable has a cumulative distribution function, whereas only discrete random variables have a probability mass function.

2.2 Computing probabilities using the cdf

Suppose X is a random variable and F_X is the distribution function for X.

1. By *definition* of F_X, we have $\mathbb{P}(X \leqslant b) = F_X(b) \ \ \forall b \in \mathbb{R}$.

2. Suppose $a < b$, then

$$\mathbb{P}(a < X \leqslant b) = \mathbb{P}(X \leqslant b) - \mathbb{P}(X \leqslant a) = F_X(b) - F_X(a)$$

   by definition of F_X.

3. Suppose $b \in \mathbb{R}$. Then the Continuity Property of the probability measure $\mathbb{P}$ implies that

$$\boxed{\mathbb{P}(X < b) = \lim_{x \uparrow b} F_X(x)}$$

4. For any $b \in \mathbb{R}$:

$$\begin{aligned}\mathbb{P}(X = b) &= \mathbb{P}(X \leqslant b) - \mathbb{P}(X < b) \\ &= F_X(b) - \lim_{x \uparrow b} F_X(x) \\ &= \underline{\text{size of the jump}} \text{ in } \ F_X(x) \text{ at } x = b\end{aligned}$$

   Remark: If the df F_X is a continuous function, then $\lim_{x \uparrow b} F_X(x) = F_X(b)$, and it follows that when F_X is a continuous function, we have

   $\forall x \in \mathbb{R} : \ \mathbb{P}(X = x) = 0.$

5. Suppose $a < b$, then

$$\mathbb{P}(a < X < b) = \lim_{x \uparrow b} F_X(x) - F_X(a)$$
$$\mathbb{P}(a \leqslant X \leqslant b) = F_X(b) - \lim_{x \uparrow a} F_X(x)$$

   Remark: When F_X is a continuous function, we have

   $\mathbb{P}(a \leqslant X \leqslant b) = \mathbb{P}(a < X \leqslant b) = \mathbb{P}(a < X < b) = \mathbb{P}(a \leqslant X < b) = F_X(b) - F_X(a).$

Example 2.6 (Example 2.5 continued.)**.** Recall Example 2.5: the random variable X denotes the number of Heads in 3 flips of a fair coin. Then X has cumulative distribution function, $F_X(x)$, given by

$$F_X(x) = \mathbb{P}(X \leqslant x) = \begin{cases} 0 & x < 0 \\ p_X(0) & 0 \leqslant x < 1 \\ p_X(0) + p_X(1) & 1 \leqslant x < 2 \\ p_X(0) + p_X(1) + p_X(2) & 2 \leqslant x < 3 \\ 1 & x \geqslant 3 \end{cases} = \begin{cases} 0 & x < 0 \\ 1/8 & 0 \leqslant x < 1 \\ 1/2 & 1 \leqslant x < 2 \\ 7/8 & 2 \leqslant x < 3 \\ 1 & x \geqslant 3 \end{cases}$$

The cumulative distribution function F_X is a step function.

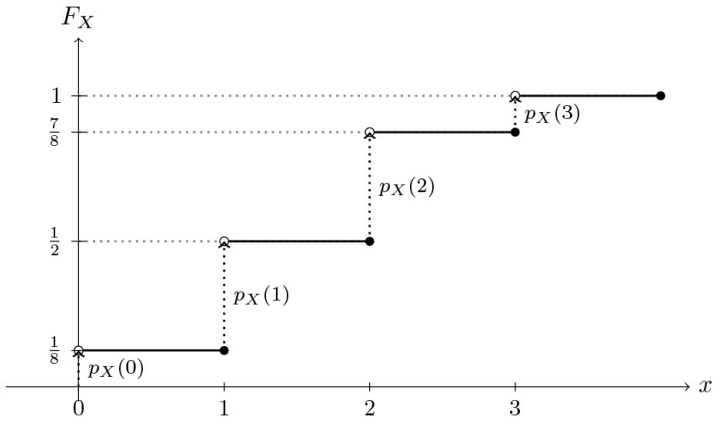


Figure 2.1: Sketch of $F_X(x)$

Given the cumulative distribution function, F_X, we can calculate probabilities:
From Property 4, we obtain: $\mathbb{P}(X = 1) = F_X(1) - \lim_{x\uparrow 1} F_X(x) = 1/2 - 1/8 = 3/8$.
From Property 2, we obtain: $\mathbb{P}(5/4 < X \leqslant 3/2) = F_X(3/2) - F_X(5/4) = 1/2 - 1/2 = 0$.
(This is as expected, since X only takes the values $\{0, 1, 2, 3\}$ and none of these values are contained in the interval $(5/4, 3/2]$, so $\mathbb{P}(5/4 < X \leqslant 3/2) = 0$.)
Remark: If X is a discrete random variable, then F_X is a step function. If F_X has a step at $x = b$, then the size of the step at $x = b$ equals $\mathbb{P}(X = b)$. All of the information about X's distribution is 'stored' in the sizes of the jumps in the cdf F_X. Therefore, it is just as easy to work with the probability mass function p_X (which also describes the distribution of X) as it is to work with F_X.

2.3 Properties of distribution functions

What properties do *all* cumulative distribution functions have?

Suppose that F_X is a cumulative distribution function for a random variable X. Then:

1. $\forall x \in \mathbb{R},\ 0 \leqslant F_X(x) \leqslant 1$

2. If $x < y$ then $F_X(x) \leqslant F_X(y)$, i.e. F_X is a non-decreasing function.

3. $\lim_{x\uparrow\infty} F_X(x) = 1$ and $\lim_{x\downarrow-\infty} F_X(x) = 0$.

4. The cumulative distribution function F_X is right continuous - this means that $\forall b \in \mathbb{R},\ \lim_{x\downarrow b} F_X(x) = F_X(b)$ and the cumulative distribution function F_X has left limits - this means that $\forall b \in \mathbb{R},\ \lim_{x\uparrow b} F_X(x)$ *exists*, but limit might not be equal to $F_X(b)$.

Proof:
1. This follows from the definition of $F_X(x) = \mathbb{P}(X \leqslant x)$ and since $0 \leqslant \mathbb{P}(X \leqslant x) \leqslant 1$.
2. Since $x < y$, we have $\{X \leqslant x\} \subseteq \{X \leqslant y\rangle$. So the monotone property of probability measures $\Rightarrow F_X(x) = \mathbb{P}(X \leqslant x) \leqslant \mathbb{P}(X \leqslant y) = F_X(y)$.
3. This follows from the definition of F_X and the continuity of probability measures.
4. This follows from the definition of F_X and the continuity of probability measures.

Any function that satisfies the above 4 properties is a distribution function for some random variable. So, to decide whether a function is a valid distribution function - sketch it and check whether the above 4 properties are satisfied.

2.4 Independent random variables

The definition of independent random variables is a natural extension of the definition of independent events.

Definition
The random variables $X_1, X_2, \ldots X_n$ are independent if and only if for all subsets $A_1, A_2, \ldots, A_n \subseteq (-\infty, \infty)$, we have

$$\mathbb{P}(X_1 \in A_1, X_2 \in A_2, \ldots, X_n \in A_n) = \mathbb{P}(X_1 \in A_1)\mathbb{P}(X_2 \in A_2)\ldots\mathbb{P}(X_n \in A_n).$$

Remarks

1. The above definition applies to both discrete and non-discrete random variables.
2. In practice, if the random variables $X_1, X_2, \ldots, X_n$ depend on physically independent 'subexperiments' then we make the assumption that the random variables $X_1, X_2, \ldots, X_n$ are (statistically) *independent* r.v.s.
   Example. Suppose that we roll a die twice. Then each roll is a 'physically independent' subexperiment — i.e. the outcome on the first roll doesn't physically influence the outcome on the second roll. So if X_1 is the outcome on the first roll and X_2 is the outcome on the second roll, we will *assume* that the variables X_1 and X_2 are (statistically) independent because the rolls are 'physically independent'.
3. If $X_1, X_2, \ldots, X_n$ are independent random variables then so are the random variables $g_1(X_1), g_2(X_2), \ldots, g_n(X_n)$ for any functions $g_1, g_2, \ldots, g_n$. This is an important result which will be used many times in this course and in other courses.

Example 2.7. Suppose a *fair* die is rolled twice. Let X_1 be the score on the first roll, and X_2 the score on the second roll. Assuming that X_1, X_2 are independent random variables, what is the probability that both scores are less than 4?

Answer: For $i = 1, 2$, each X_i has probability mass function p_{X_i} given by

$$p_{X_i}(x) = \begin{cases} 1/6 & x = 1 \\ 1/6 & x = 2 \\ 1/6 & x = 3 \\ 1/6 & x = 4 \\ 1/6 & x = 5 \\ 1/6 & x = 6 \end{cases}$$

The required probability is

$$\begin{aligned} \mathbb{P}(X_1 < 4, X_2 < 4) &= \mathbb{P}(X_1 \leqslant 3, X_2 \leqslant 3) \\ &= \mathbb{P}(X_1 \leqslant 3)\mathbb{P}(X_2 \leqslant 3) \qquad \text{(by independence of } X_1, X_2) \\ &= (3/6)(3/6) = 1/4 \end{aligned}$$

2.5 Common examples of discrete random variables

In this section we will go through a range of important distributions which will regularly appear in probability and statistics applications. Some of these you may have seen before, here we will be explaining them in the context of this chapter. Note that you will find a summary of these

distributions in the "Formula Sheet For Probability and Statistics Exams" available on Canvas and provided in the exam.

2.5.1 Bernoulli distribution

Bernoulli variables are special, simple random variables which only take two possible values: 0 or 1. We say that the random variable X is a Bernoulli random variable if and only if $\mathrm{Im}X = \{0, 1\}$.

Examples

1. Flip a coin. The sample space is $\Omega = \{\mathrm{H}, \mathrm{T}\}$ and we can define the Bernoulli random variable $X : \Omega \to \mathbb{R}$ by $X(\mathrm{H}) = 1$ and $X(\mathrm{T}) = 0$.
2. Turn on a light and measure the lifetime (in hours) of the light bulb. The sample space is $\Omega = [0, \infty)$. If we are interested in whether the light bulb lasts more than 20 hours, we can define the Bernoulli random variable $Y : \Omega \to \mathbb{R}$ by $Y(\omega) = 1$ if the lifetime $\omega \geq 20$, and $Y(\omega) = 0$ if the lifetime $\omega < 20$. So the random variable Y answers the simple Yes/No question: 'Is the lifetime greater than 20 hours?', where the value 1 corresponds to 'Yes' and the value 0 corresponds to 'No'.

Properties of Bernoulli random variables
Suppose X is a Bernoulli random variable. Then we must have:

$$\mathbb{P}(X = 1) = p \quad \text{for some} \quad 0 \leqslant p \leqslant 1$$
$$\textit{and} \quad \mathbb{P}(X = 0) = 1 - p$$

Definition
We define the **indicator variable** $\mathbb{I}_A$ for the event A by

$$\mathbb{I}_A = \begin{cases} 1 & \text{if event } A \text{ happens} \\ 0 & \text{if event } A \text{ does not happen} \end{cases}$$

Typically, Bernoulli variables arise as **indicator variables** for events of interest.

2.5.2 Binomial distribution

Suppose we have a sequence of n independent, identical, subexperiments (often called 'trials'), each having two possible outcomes, 'Success' or 'Failure', with
$\mathbb{P}(\text{'Success'}) = p$ and $\mathbb{P}(\text{'Failure'}) = 1 - p$.

Let $X =$ total number of successes in the n trials.

Then we say that X is a Binomial random variable with parameters n (number of trials) and p (probability of success in a trial) and we write $X \sim \mathrm{Bin}(n, p)$.

It is clear from the definition of X that $\mathrm{Im}X = \{0, 1, 2, \ldots, n\}$.

The probability mass function for $X \sim \mathrm{Bin}(n, p)$ is given by

$$\boxed{p_X(k) = \mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \text{ for } k = 0, 1, 2, \ldots n.}$$

Explanation:

$\binom{n}{k}$ = # ways to choose a subset of size k from $\{1, 2, \ldots, n\}$
= # ways to position k successes in a sequence of k successes and $(n-k)$ failures.

Also, each different sequence of k successes and $(n-k)$ failures has probability $p^k(1-p)^{n-k}$ since the trials are independent so we can multiply the probabilities of successes and failures together.

Example 2.8. Flip a *biased* coin 3 times and suppose that $\mathbb{P}(\text{'Heads'}) = 1/3$ and $\mathbb{P}(\text{'Tails'}) = 2/3$. Let X = number of Heads in 3 flips.
Then $X \sim \text{Bin}(3, \frac{1}{3})$, Im$X$=$\{0, 1, 2, 3\}$ and the probability mass function for X is given by:

$$p_X(0) = \mathbb{P}(X=0) = \binom{3}{0}\left(\frac{1}{3}\right)^0\left(\frac{2}{3}\right)^3 = \frac{8}{27}$$
$$p_X(1) = \mathbb{P}(X=1) = \binom{3}{1}\left(\frac{1}{3}\right)^1\left(\frac{2}{3}\right)^2 = \frac{12}{27}$$
$$p_X(2) = \mathbb{P}(X=2) = \binom{3}{2}\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^1 = \frac{6}{27}$$
$$p_X(3) = \mathbb{P}(X=3) = \binom{3}{3}\left(\frac{1}{3}\right)^3\left(\frac{2}{3}\right)^0 = \frac{1}{27}$$

Representation in terms of indicator variables
If we define the indicator random variables $\mathbb{I}_k$ for $k = 1, 2, \ldots, n$ to be

$$\mathbb{I}_k = \begin{cases} 1 & \text{if trial } k \text{ is a Success} \\ 0 & \text{if trial } k \text{ is a Failure} \end{cases}$$

then we can represent $X \sim \text{Bin}(n, p)$ as $X = \mathbb{I}_1 + \mathbb{I}_2 + \cdots + \mathbb{I}_n$

2.5.3 Geometric distribution

Suppose now that we have an *infinite* sequence of independent, identical trials and each trial has only two possible outcomes, 'Success' or 'Failure', with $\mathbb{P}(\text{'Success'}) = p$ and $\mathbb{P}(\text{'Failure'}) = 1-p$.

Let Y = number of trials until the first success is obtained.

Then we say that Y is a Geometric random variable with parameter p and we write $Y \sim \text{Geo}(p)$. It is clear from the definition of Y that ImY=$\{1, 2, 3, \ldots\}$. The event $\{Y = y\}$ occurs when the first $(y-1)$ trials are all Failures, and then trial y is a Success. Using the independence of the trials, the probability of this is $(1-p)^{y-1}p$.
That is, the probability mass function for $Y \sim \text{Geo}(p)$ is given by

$$\boxed{p_Y(y) = \mathbb{P}(Y=y) = (1-p)^{y-1}p \quad \text{for } y = 1, 2, 3, \ldots}$$

Example 2.9. Flip a *biased* coin repeatedly until the first flip to land 'Heads', and suppose that for each flip, $\mathbb{P}(\text{'Heads'}) = 1/3$ and $\mathbb{P}(\text{'Tails'}) = 2/3$. Find the probability that more than 2 flips are needed.

Answer: Denoting by Y the number of flips needed, then $Y \sim \text{Geo}(1/3)$, with probability mass function

$$p_Y(y) = \mathbb{P}(Y=y) = (2/3)^{y-1}\,1/3 \quad \text{for } y = 1, 2, 3, \ldots$$

We are asked to find $\mathbb{P}(Y > 2)$:

$$\begin{aligned}\mathbb{P}(Y > 2) &= 1 - \mathbb{P}(Y \leq 2) \\ &= 1 - \mathbb{P}(Y = 2) - \mathbb{P}(Y = 1) \\ &= 1 - (2/3)\,(1/3) - 1/3 = 4/9 \approx 0.444\end{aligned}$$

Example 2.10. Products produced by a machine have a 20% defective rate. What is the probability that the first defective product occurs in the first five inspections? Then, find the average number of inspections to obtain the first defective product.

Answer: Let X be the number of inspections to obtain the first defective product, then $X \sim \text{Geo}\,(0.2)$. Then the probability that the first defective product occurs within the first five inspections is

$$\begin{aligned}\mathbb{P}(X \leq 5) &= \mathbb{P}(X = 1) + \mathbb{P}(X = 2) + \mathbb{P}(X = 3) + \mathbb{P}(X = 4) + \mathbb{P}(X = 5) \\ &= 0.20 + (0.80)(0.20) + (0.80)^2(0.20) + (0.80)^3(0.20) + (0.80)^4(0.20) \\ &= 0.67232.\end{aligned}$$

and the expected number of inspections to find the first defective product is

$$\mathbb{E}(X) = \frac{1}{0.20} = 5.$$

2.5.4 Poisson distribution

The Poisson distribution is often used as a model for data describing the number of events of a particular kind that occur in a given time interval, for example:

- the number of claims on a portfolio of policies in a year;
- the number of flooding events that might occur on a certain river over a 5-year period;
- the number of industrial accidents that occur in a factory in a year.

Definition 2.5. Suppose $\lambda > 0$ is a fixed constant. We say that the random variable X has a Poisson distribution with parameter λ, denoted $X \sim \text{Poi}(\lambda)$, if $\text{Im}X = \{0, 1, 2, \dots\}$ and

$$\boxed{p_X(x) = \mathbb{P}(X = x) = \frac{\lambda^x e^{-\lambda}}{x!} \qquad \text{for } x = 0, 1, 2, \dots}$$

We will show in Section 8.4 that the Poisson distribution arises as the limit of a Binomial distribution.

Example 2.11. The number of flooding events that occur on a certain river over a 5 year period is believe to follow a Poisson distribution with parameter $\lambda = 4$. Find the probability that the number of flooding events over a 5 year period is 2 or less.

Answer: Denoting by X the number of flooding events over a 5 year period, then $X \sim \text{Poi}(4)$, with probability mass function

$$p_X(x) = \mathbb{P}(X = x) = \frac{4^x e^{-4}}{x!} \qquad \text{for } x = 0, 1, 2, \dots$$

We are asked to find $\mathbb{P}(X \leqslant 2)$:

$$\begin{aligned}\mathbb{P}(X \leqslant 2) &= \mathbb{P}(X = 0) + \mathbb{P}(X = 1) + \mathbb{P}(X = 2) \\ &= \frac{4^0 e^{-4}}{0!} + \frac{4^1 e^{-4}}{1!} + \frac{4^2 e^{-4}}{2!} \\ &= 13e^{-4} \approx 0.238\end{aligned}$$